Internship Title: 2025

Internship Program Name: Cyber Security

Your Name: Areesh Awan

Internship Lead Name: Faizyab

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H.12, Islamabad, Pakistan

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Executive Summary

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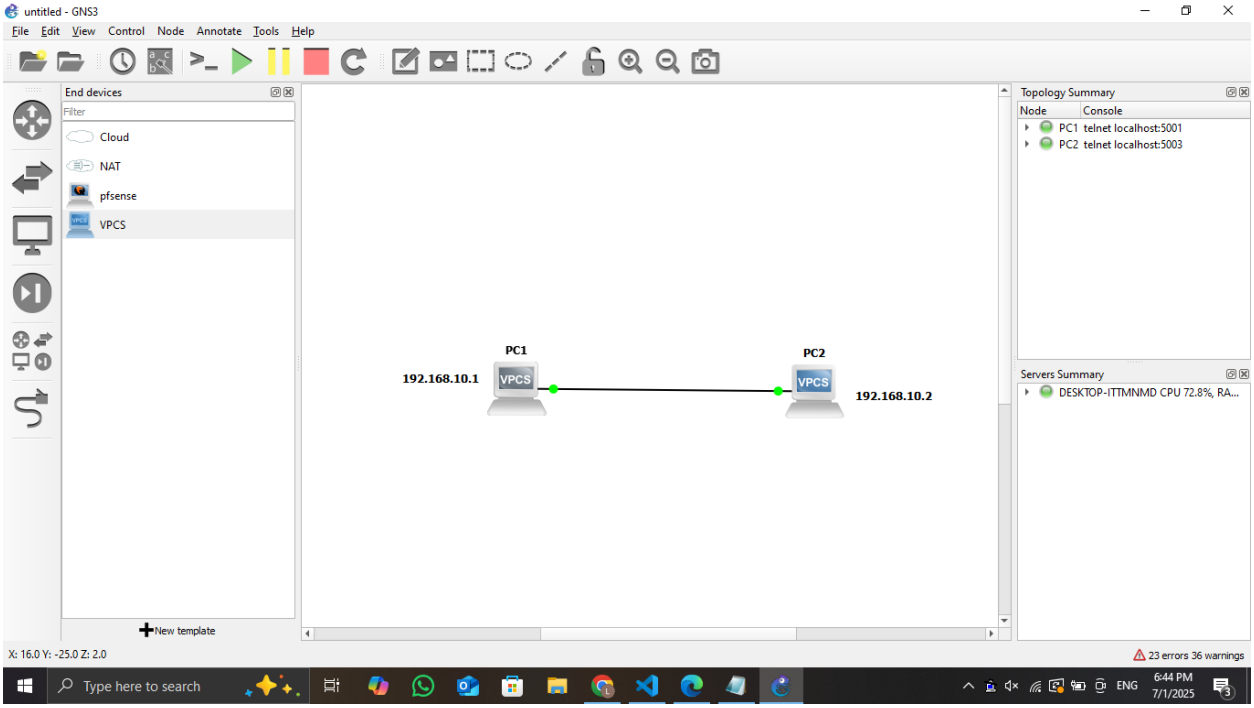
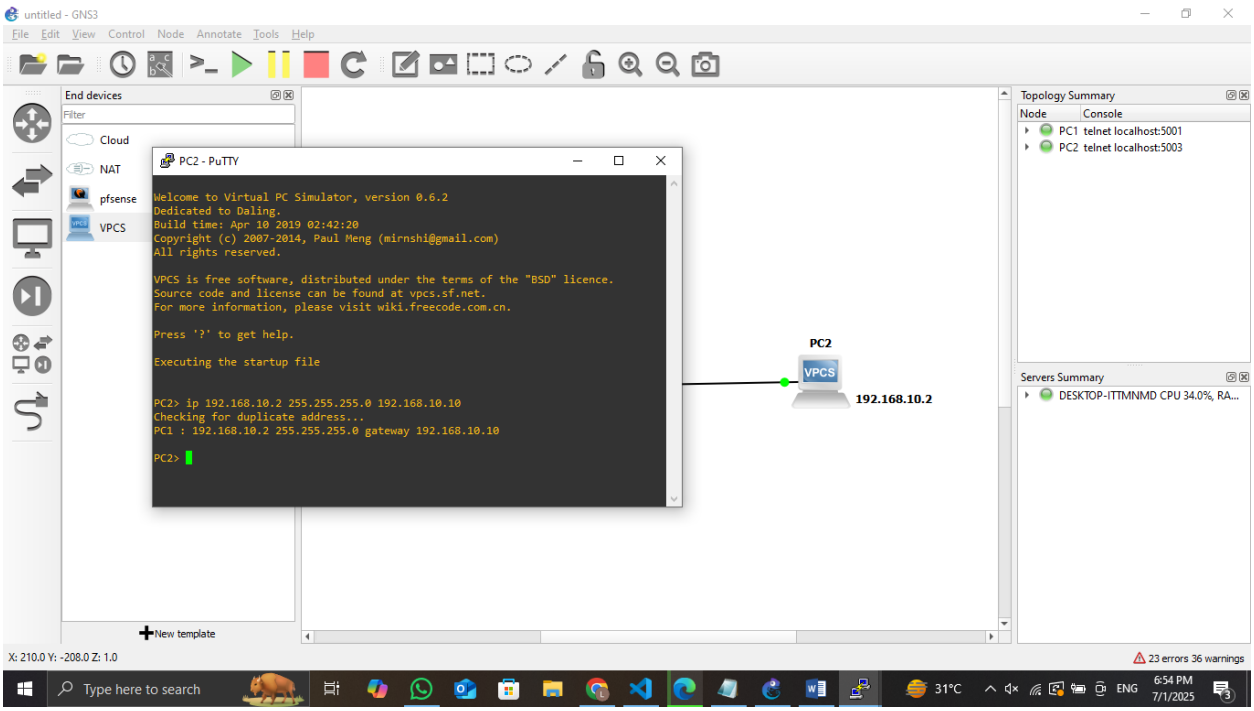
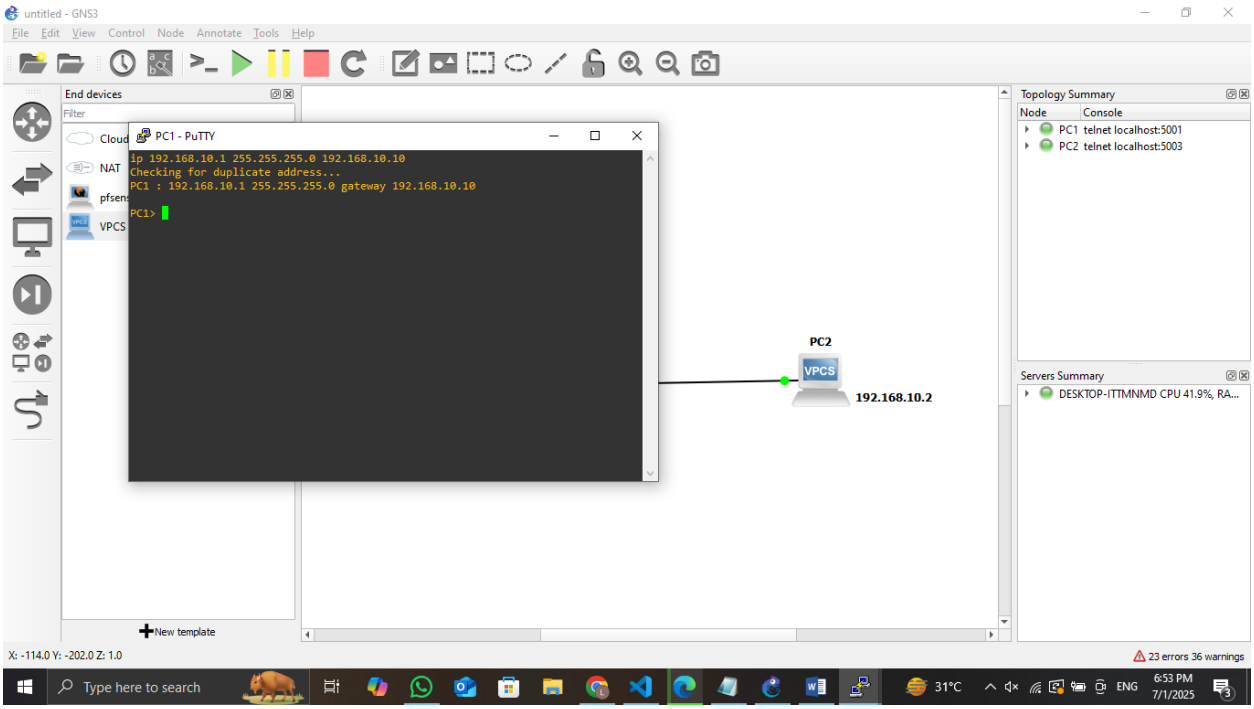
Step 1

Drag and connect

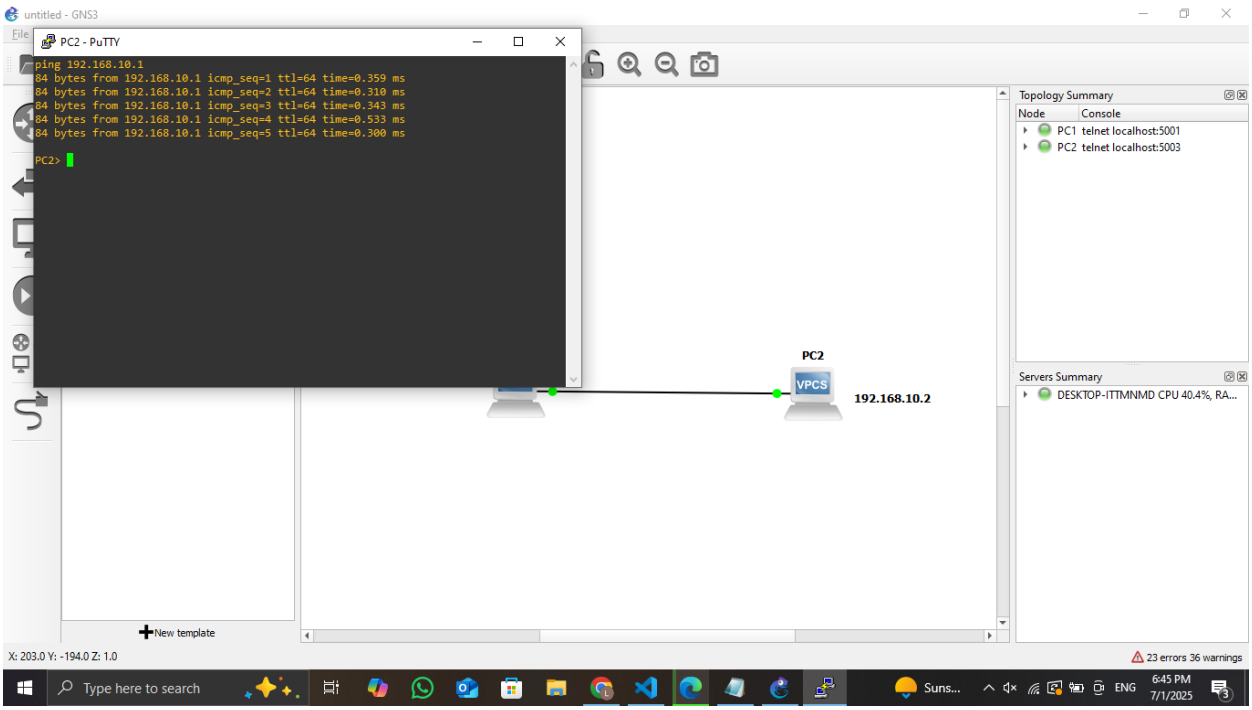






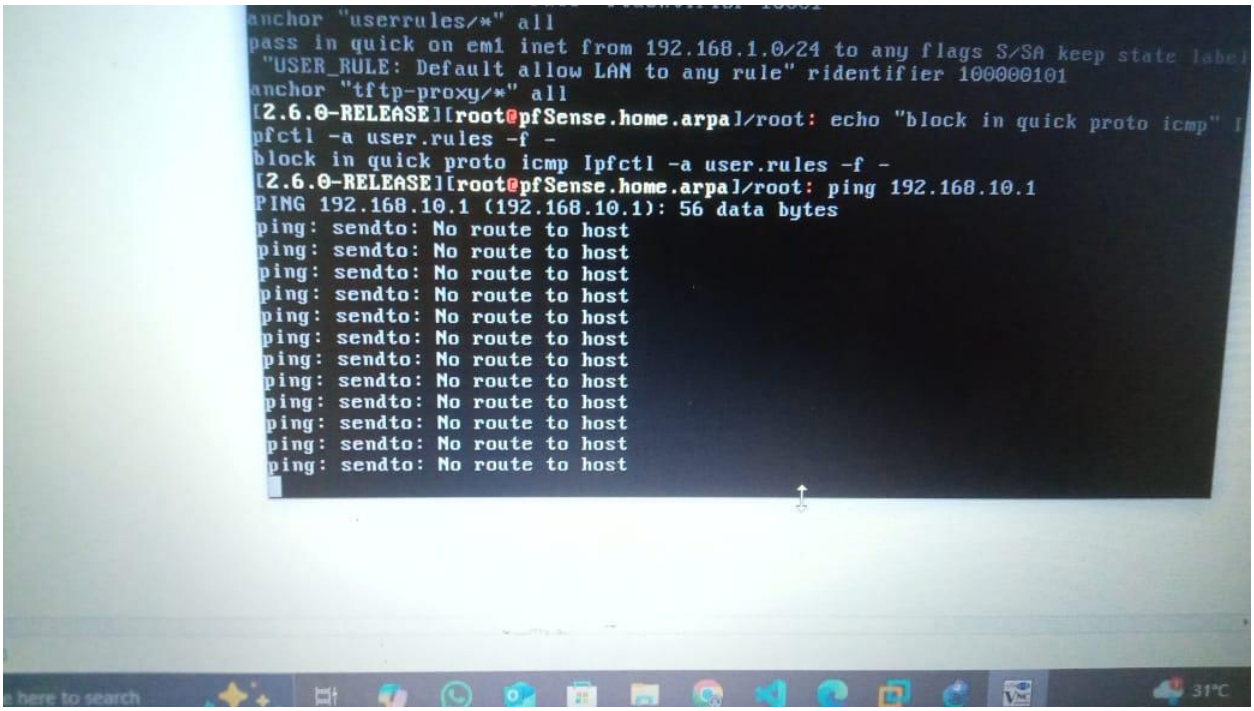
Step 2

Ping test



Step 3

Block ping


anchor "userrules/*" all
pass in quick on em1 inet from 192.168.1.0/24 to any flags S/SA keep state label
"USER_RULE: Default allow LAN to any rule" ridentifier 100000101
anchor "tftp-proxy/*" all
[2.6.0-RELEASE][root@pfSense.home.arpa]/root: echo "block in quick proto icmp" |
pfctl -a user.rules -f -
block in quick proto icmp Ipfctl -a user.rules -f -
[2.6.0-RELEASE][root@pfSense.home.arpa]/root: ping 192.168.10.1
PING 192.168.10.1 (192.168.10.1): 56 data bytes
ping: sendto: No route to host
ping: sendto: No route to host
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ping: sendto: No route to host
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ifier 1000002663
pass out on lo0 inet6 all flags S/SA keep state label "pass IPv6 loopback" ri
tifier 1000002664
pass out inet all flags S/SA keep state allow-opts label "let out anything IP
from firewall host itself" ridentifier 1000002665
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pass in quick on em1 proto tcp from any to (em1) port = https flags S/SA keep
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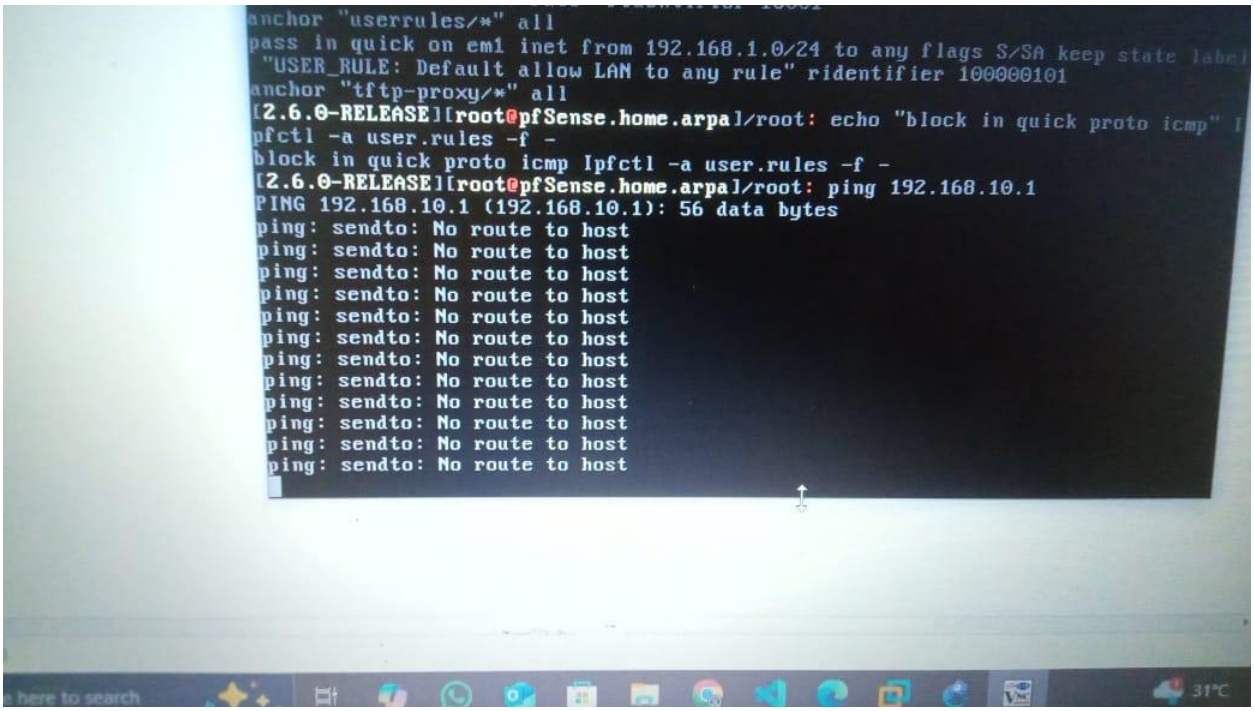
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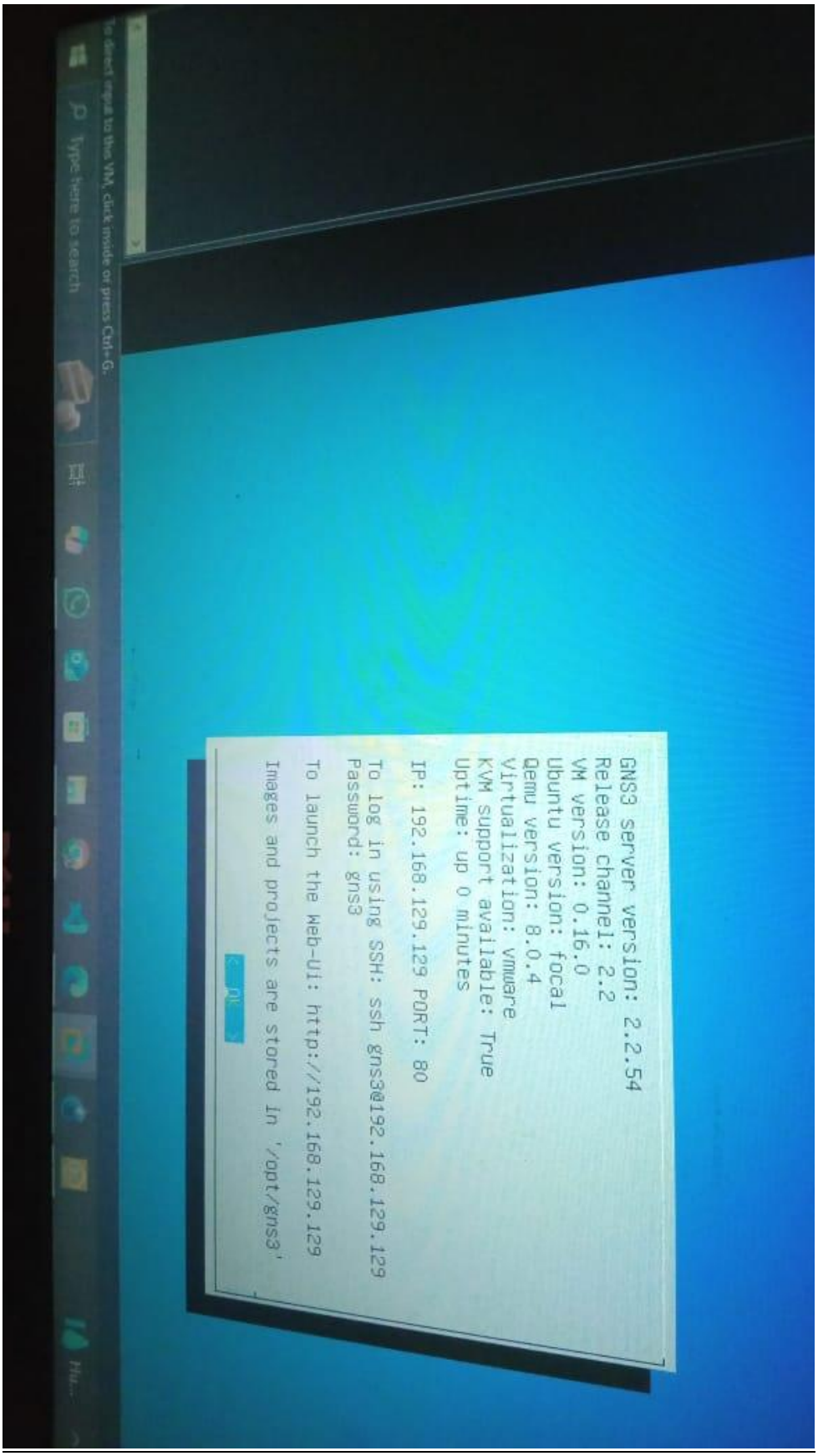
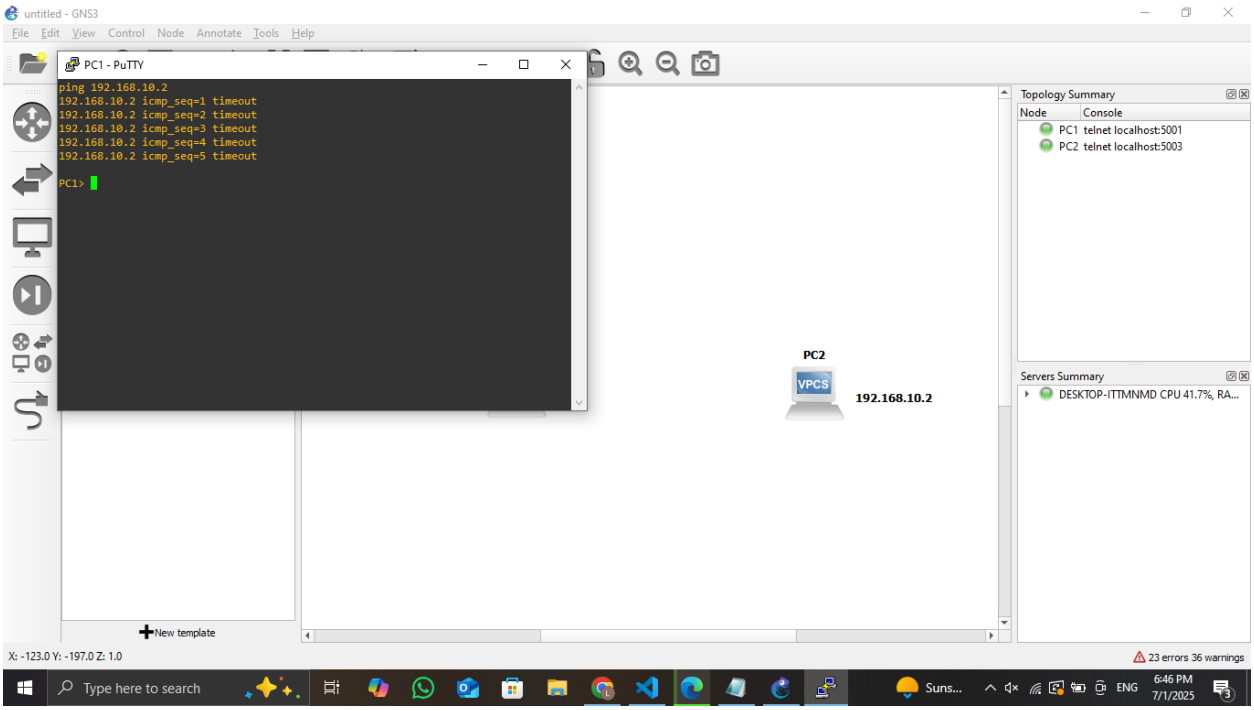

GNS3 server version: 2.2.54
Release channel: 2.2
VM version: 0.16.0
Ubuntu version: focal
Qemu version: 8.0.4
Virtualization: vmware
KVM support available: True
Uptime: up 0 minutes
IP: 192.168.129.129 PORT: 80
To log in using SSH: ssh gns3@192.168.129.129
Password: gns3
To launch the Web-UI: http://192.168.129.129
Images and projects are stored in '/opt/gns3'
OK
To direct input to this VM, click inside or press Ctrl+G.
Type here to search



Q1: What does ping show?

Ping is a command used to check if one device can connect to another device over a network. It shows if the other device is online and how long it takes for a small message to travel to that device and come back. If the connection is working, you will see a reply with the time it took in milliseconds. If the connection is not working, you may see a message like “Request timed out.” Ping helps test network connection and find problems.

Q2:Why is it important to block unwanted pings?

It is important to block unwanted pings to keep your network safe. If someone sends many pings to your system, it can slow down your network or even crash it. Hackers may use pings to find active devices or attack them. Blocking unwanted pings helps hide your devices and protects them from attacks. It is a simple way to improve security and reduce the risk of being targeted.

Skilled Learned

Q:1What is ping and how it tests connectivity?

Ping is a network tool used to check if one computer can talk to another over a network. It works by sending a small message to the other device and waiting for a reply. If the reply comes back, it means the connection is working. Ping also shows how fast the message travels, which helps measure the speed and delay in the network. If there is no reply, it means the device is not reachable or there is a problem in the network.

Q2: What happens when firewall blocks traffic?

When a firewall blocks traffic, it stops unwanted or unsafe data from entering or leaving a computer or network. This means that the blocked traffic cannot reach its destination. For example, if someone tries to access a website or send data, and the firewall sees it as unsafe, it will block that action. As a result, the connection fails, and the user cannot reach the website or service. This helps protect the system from hackers, viruses, and other threats.

Q3: How to use GNS3 to simulate and observe security behavior visually?

GNS3 is a powerful network simulator that allows us to connect virtual devices like routers, firewalls, and PCs. By using it, we can create different network setups and apply security rules such as firewall rules, blocking ping, and testing attacks. When we connect devices in GNS3, we can see how the data flows and where it gets blocked. This tool is very helpful when we want to test network attacks and responses safely, without harming a real network.

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